

SABARAGAMUWA UNIVERSITY OF SRILANKA
FACULTY OF COMPUTING
DEPARTMENT OF COMPUTING AND INFORMATION SYSTEMS
IS4103 – OS(P)
ASSIGNMENT 02
DURATION- 30 MINUTES

Scenario

In high-availability systems, a "Watchdog" is a process that monitors workers. If a worker process hangs up, freezes, or takes too long to respond, the watchdog process must intervene and forcefully terminate the worker to release trapped hardware resources.

Requirements

Write a C program that mimics this infrastructure using signal handling mechanisms:

1. The main program forks into a Parent (acting as the Watchdog) and a Child (acting as the Worker).
2. The Child Worker process should print its PID and state: `[Worker] Initialized. Beginning heavy computation....` It must then attempt to sleep for 100 seconds to simulate an intensive computation loop or a system hang.
3. The Parent Watchdog process should print: `[Watchdog] Monitoring worker... Allocation limit: 3 seconds.`
4. The Watchdog must sleep for exactly 3 seconds, simulating the maximum allowed time slice for the worker.
5. After waking up, if the child is still running, the Watchdog must forcefully terminate the child process system call with the `SIGKILL` signal.
6. The parent must then cleanly clear the terminated child from the system process table and print: `[Watchdog] Worker exceeded time limit. Forced termination complete.`